A 65nm CMOS 1.2V 12b 30MS/s ADC with Capacitive Reference Scaling

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***Abstract*- A 1.2V 12b 30MS/s pipelined ADC, implemented in a 65nm standard CMOS technology, achieves an SNDR of 65.1dB with a rail-to-rail 4.7MHz input. A capacitive reference scaling technique is proposed to alleviate the high gain requirement of the opamp and a wide input range of 2.4Vp-p differential for low voltage operation in the nanometer domain. The prototype ADC dissipates 18mW and occupies an active die area of 0.34mm². The measured DNL and INL are ±0.44LSB and ±1.33LSB, respectively.**

I. INTRODUCTION

As the processes move into nanometer scale era, the analog performance has been degraded due to lowered output impedance of transistors, reduced input signal power from the scaled supply voltage, and increased on-resistance of switches. Especially, the continuous degradation of opamp gain with the process development is one of the most substantial difficulties in low voltage high resolution ADC designs for the mobile communication systems. Recently, sub-ranging architecture without closed-loop or 3-stage opamp are suggested to avoid or to meet the high gain requirement [1-2]. In this paper, a capacitive reference scaling (CRS) technique is proposed as a viable solution to lower opamp gain requirement in the Multiplying D/A Converter (MDAC), to enable rail-to-rail wide input dynamic range for better SNR, and to improve settling behavior of MDACs from better reference sampling for low voltage low power operation.

II. OP AMP GAIN REQUIREMENT IN MDACS

The minimum opamp gain requirement in the conventional MDACs of pipeline ADCs is derived from [3] as

$$\frac{2^k}{1+(1+2^k+\alpha_P)/A_O} \geq 2^k(1-2^{-(N-m)}) \qquad (1)$$

where 2^k is the interstage gain, Ao is a finite gain of non-ideal opamp, α_p is a ratio of summing node parasitic capacitance to feedback capacitance, 2^N is the accuracy of input, and m is the stage resolution. Neglecting α_p, the required gain of opamp is almost same with the accuracy of input even with large m as long as k=m. On the other hand, the gain requirement of opamp can be reduced by 6×(m-k) dB by designing m larger than k. Instead, the output dynamic range is reduced by 2×(m-k) from the input range with the decreased stage resolution of the next. Also, it does not degrade the noise budget for the next stage because the required accuracy of the reduced output range is relaxed as well. The disadvantage is that the next stage demands another reference voltage for the reduced dynamic range.

III. PROPOSED ADC ARCHITECTURE

The proposed 1.2V 12b 30MS/s ADC without a dedicated front-end SHA consists of a 3b first stage followed by three 2.5b stages terminated by a 3.5b flash ADC, along with other supplementary blocks as shown in Fig. 1. The Nyquist sampling is achieved by the use of boot-strapped switch maintaining low on-resistance and by careful layout reducing sampling aperture error of the first stage input path.

The proposed CRS dealing with the different input ranges is depicted in the bottom of Fig. 1. In the first stage, k=2 and m=3 is chosen so that α_p is far less than $1+2^k$ and the gain requirement of opamp for 12 bit resolution takes advantage of reduction of 6dB according to (1). Therefore, the rail-to-rail input is converted to 3-bit digital codes by 7 comparators across the reference voltage. At the same time, the residue is multiplied by 2^2 to meet the gain requirement as well as output dynamic range with headroom margin of the opamp in MDAC1. On the other hand, in the remaining MDACs, the halved input range is converted to 3 bit digital codes including a bit for digital correction by 5 comparators while the residue amplification ratio is same as 2^2. The last stage, FLASH5, consists of 9 comparators with 1 bit interpolation for 12b outputs. All MSBs of output codes corresponding to each comparator from the second stage to the last are used for the over-range detection.

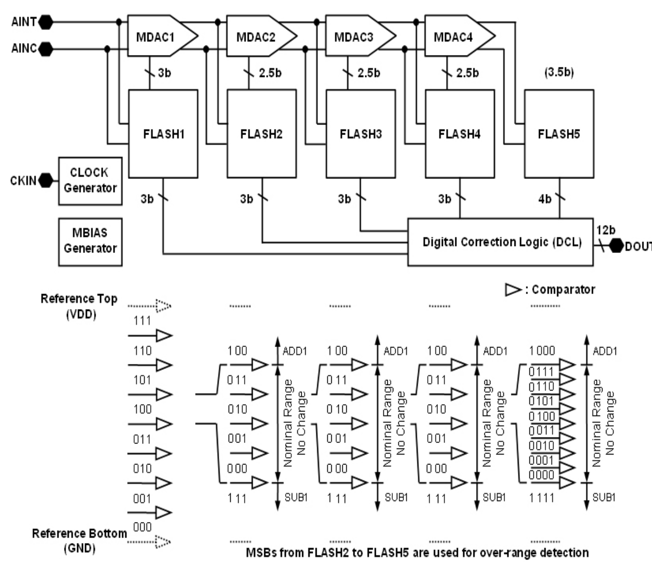

Fig. 1. Block diagram of the proposed ADC with coding technique.

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In a subranging flash ADC, comparators are typically implemented along a resistor string in series with same voltage interval suffering from same amount of static and dynamic offsets. Digital correction logic (DCL) can adjust the error from comparator offsets by overlapping one bit of the output digital codes from each stage. In conventional designs, a half interval between decision levels decided by comparators is multiplied and overlapped in the next stage for the correction. In this work, the correction range is designed as 1/12 of the interval, but it has advantage of doubled input dynamic range of conventional one. After all, the effective correction voltage range is one third compared to that of the classical scheme. For the FLASH sub ADCs with low offset margin, the comparator must be designed and laid out carefully. The transitions of all node voltage levels of input pair in comparators, especially, are designed to be minimized so that kick-back noise is reduced thereby lessening the influence on analog signal and reference voltages. From this, the low offset requirement can be satisfied by using power efficient latch-only structure. Therefore, it can save additional stages from more offset margin for the reduction of size and power consumption.

With the CRS, the same reference voltage is applied to the halved input range by capacitive scaling instead of additional appropriate reference selection [4]. In this design, the power supply (VDD) and ground (GND) are used for reference top and bottom, respectively, to increase signal power by rail-to-rail input and to improve reference sampling by setting VDD as gate-source voltage of reference switches. Also, it gets rid of reference output pins and the related elements such as external capacitors as extra benefit.

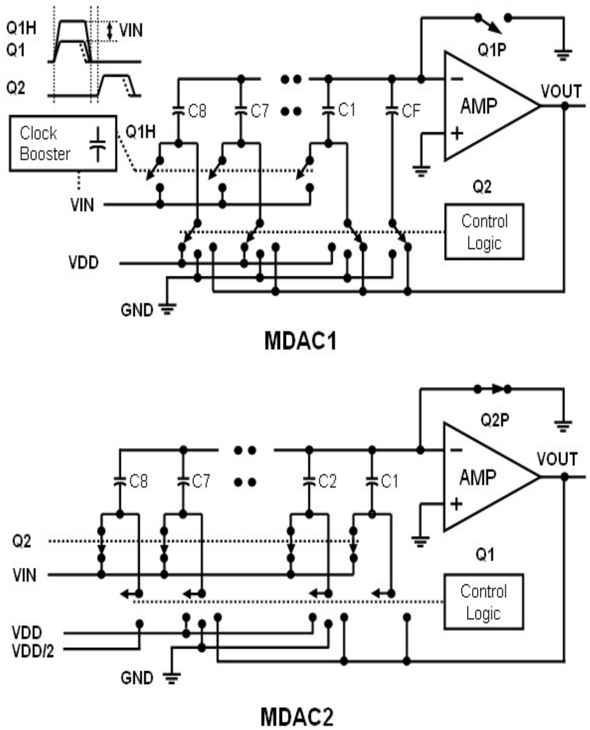

Fig. 2. Block diagrams of MDACs for the proposed CRS.

IV. Circuit Design with Capacitive Reference Scaling

The simplified block diagrams of MDAC1 and MDAC2 implementing the proposed CRS are shown in Fig. 2. The MDAC1 is composed of 9 unit capacitors, a switching block, and 2-stage opamp. It operates with two phase non-overlapping clocks denoted as Q1 and Q2; the sampling and the amplification phase in MDAC1, respectively. The Q1H generated by clock booster, in phase with Q1, has magnitude of the sum of input voltage and logic voltage of Q1. The 8 unit capacitors are used for sampling except a dedicated feedback one, CF, in the sampling phase. In amplification phase of Q2, 7 unit capacitors are connected to the reference while the selected one is used as feedback capacitor with the dedicated one to achieve interstage gain 2^2. The selection of one unit capacitor among 8 depends on the input value using a modified commutative feedback capacitor switching technique by control logic to improve linearity [5].

MDAC2 has 8 unit capacitors to scale down the reference value to half even though the number of the related comparators is 5 in the FLASH2. All the capacitors are used for sampling when Q2 is high in MDAC2. In amplification phase, Q1, the selected 5 capacitors are connected to the reference depending on the input and the other two are used as feedback like MDAC1 to maintain the interstage gain. The last unit capacitor, C8, is connected to half VDD to adjust the offset from the over-range detection bit. After all, the quantized DAC analog output corresponding to a single comparator in FLASH2 is not (2×VDD)/4 but (2×VDD)/8 like the first stage. The remaining MDACs have the same architecture as MDAC2.

A simple 2-stage folded cascode opamp using a 65nm CMOS digital process, shown in Fig. 3, is able to meet the reduced gain requirement and output dynamic range of opamp in MDAC1. The additional NMOS transistors in the first stage opamp, shown in dotted rectangle, are added and optimized to reduce the gain loss from parallel connections without extra delay. There is no loss in headroom margin from VDD gated NMOS transistors. The second stage opamp adopts a folded cascode structure of 3 CMOS stack to maintain the wide output dynamic range and to get rid of gain loss from input pair of short length. In addition, an improved compensation technique is adopted to reduce the current of opamps [6].

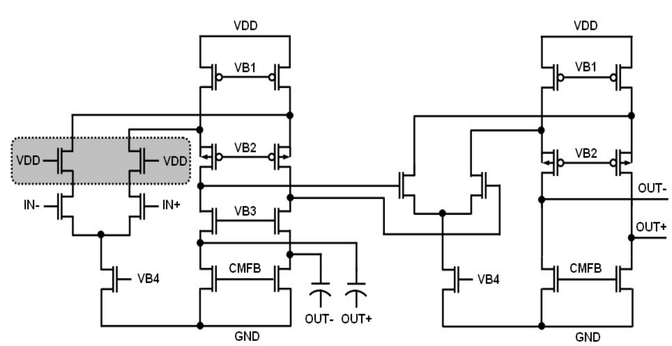

Fig. 3. Schematic of 2-stage folded cascode opamp.

V. LAYOUT CONSIDERATION

As the processes have developed, more layout issues have risen, especially related to poly gate and its channel, and they are needed to be considered for high resolution in small size.

In a subranging flash ADC, two comparators at both ends of the comparator array have relatively large static and dynamic offsets compared with the rest of the comparators due to charge-induced damages during plasma exposure. In this design, narrow poly lines connected to a fixed bias are added between the adjacent comparators and the poly gates are laid out compactly to reduce the undesirable comparator input-referred offsets instead of dummy comparators at both ends of array [7]. As a result, it provides same environment for the critical devices in comparator.

The capacitor arrays used in the entire ADC are based on Vertical Natural Capacitor (VN-Cap) using metal fringing capacitance. In this design, the capacitors in MDACs and flash ADCs consist of MET1 to MET4. They are all laid out in symmetric structure in parallel and isolated by substrate to MET5 barrier to get the same parasitic capacitance illustrated in Fig. 4. Also the employed layout technique provides the same conditions for all the unit capacitor so that it eliminates parasitic capacitance mismatches from the metal routings to get a high-resolution without extra dummy capacitor [8].

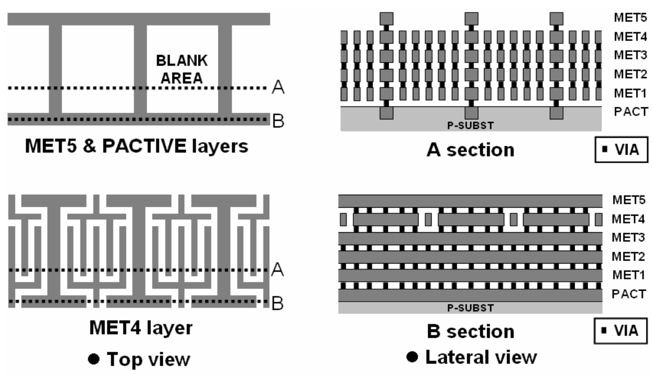

Fig. 4. Layout for high matching VN Capacitor.

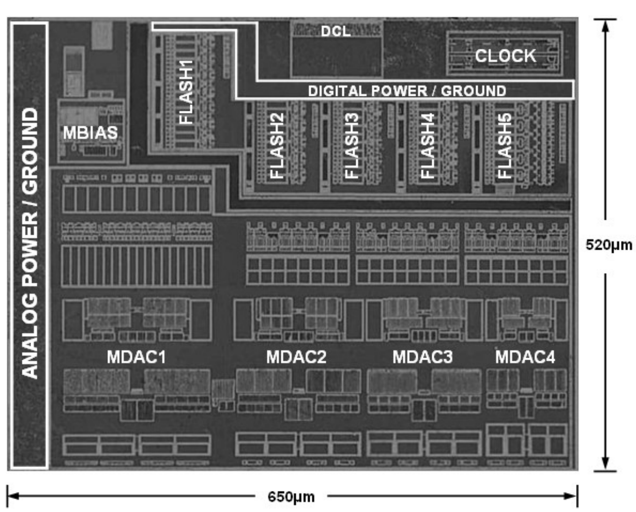

Fig. 5. Die photograph.

VI. EXPERIMENTAL RESULTS

The prototype ADC is fabricated in a 65nm single-poly five-metal standard CMOS process using no analog enhancement, and occupies 0.34mm^2 ($650\mu\text{m} \times 520\mu\text{m}$) active die area as shown in Fig. 5. There is no dummy capacitor and no dummy comparator. It dissipates 18mW at 30MS/s with a 1.2V power supply. The measured DNL and INL are ± 0.44LSB and ± 1.33LSB, respectively, as plotted in Fig. 6. The dynamic performance is measured from the FFT spectrum as shown in Fig. 7. For a 4.7MHz 2.4Vp-p differential input, the measured SNDR and SFDR are 65.1dB and 74.9dB, respectively, at a clock frequency of 30MHz. The calculated FOM, defined as Power/($2^{ENOB}\times$Fs), is 0.41pJ/Conversion-step. At the Nyquist sampling rate, the FOM is 0.46pJ/Conversion-step from the measured SNDR of 64.1dB with a 14.2MHz input.

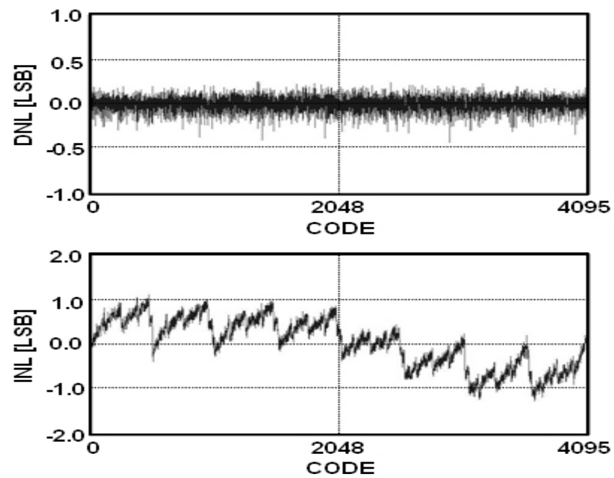

Fig. 6. Measured DNL and INL.

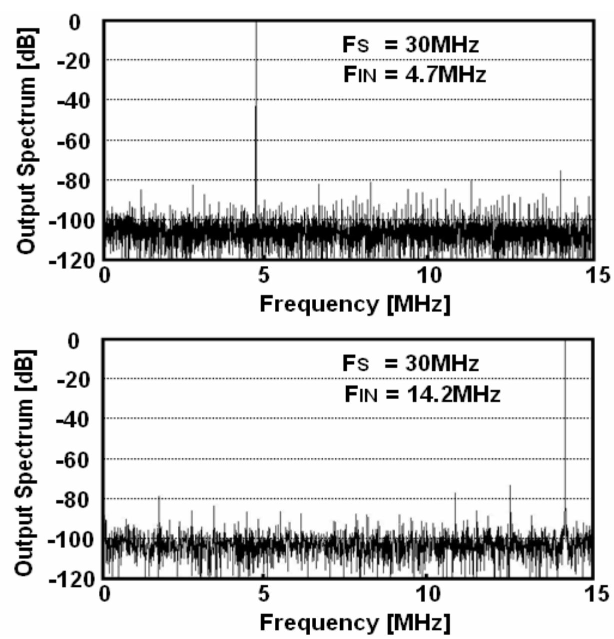

Fig. 7. Measured FFT spectrums.

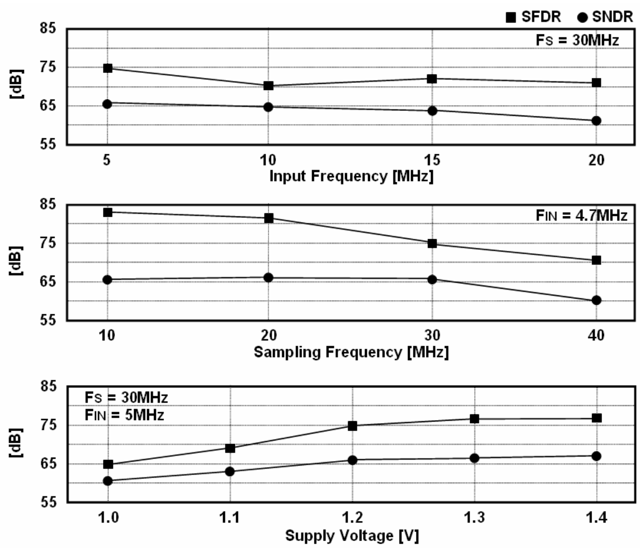

Fig. 8. SNDR and SFDR vs. input frequency, sampling frequency, and supply voltage.

The SNDR and SFDR versus the input frequency, the sampling frequency, or the supply voltage are plotted in Fig. 8. The SNDR is maintained over Nyquist sampling rate and ENOB of more than 10b can be achieved down to 1.0V power supply.

VII. CONCLUSIONS

A 65nm CMOS 1.2V 12b 30MS/s pipelined ADC has been presented. In this work, the capacitive reference scaling (CRS) technique is proposed to achieve high SNR with low gain opamp by utilizing wide input dynamic range at low supply voltage. By adopting the proposed techniques, the prototype ADC achieves over SNDR of 65dB at 1.2V supply showing FOM of less than 0.5pJ/Conversion-Step up to Nyquist sampling rate.

The overall ADC performance is summarized in table I.

Table I.
Performance Summary of the proposed ADC.

Resolution	12 bits
Conversion Rate	30 MS/s
Process	65nm 1 poly 5 metal CMOS
Supply Voltage	1.2 V
Input Range	2.4 Vp-p differential
DNL / INL	±0.44 LSB / ±1.33 LSB
SNDR/SFDR (@F_{IN} = 4.7MHz)	65.1 dB / 74.9 dB
SNDR/SFDR (@F_{IN} = 14.2MHz)	64.1 dB / 71.9 dB
Power Dissipation	18 mW
Active Die Area	0.34 mm^2 (=650µum × 520µum)

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